ARKFALL — Game Design Document

Mikel Areta Garcia u232868

Oriol Carreras Abad u232393

1. Introduction

1.1 Title

ARKFALL — *Build the beacon, bring them home.*

1.2 Abstract

Arkfall is a 3D colony-survival management game set on Tharsa, a red, dusty, Mars-like planet with thin, unbreathable air. The player leads a small landing crew who must build a colony, ferry more colonists down by rocket, and construct the Haven Beacon that will guide their damaged ark ship to a safe landing.

The player never controls a single colonist; each colonist acts on their own to keep their five needs met: Hunger, Thirst, Oxygen, Rest and Happiness. Health is tracked separately as Healthy, Injured or Dead. Resources come from mine buildings placed on deposits and staffed by Miners. More colonists arrive by rocket whenever the player calls one, but every new arrival is another mouth to feed and another body to keep breathing.

- **Genre:** colony management / survival sim
- **Platform:** PC, keyboard and mouse
- **Players:** single-player
- **Session:** one map per run, about 60–90 minutes
- **Scope for the course:** one map, 2 mined resources (iron, copper), 21 buildings (10 interior, 11 exterior), 4 jobs, 1 robot type

1.3 Lore

In 2231 the colony ark *Meridian* leaves a crowded Earth carrying 200 colonists in cryosleep. Its destination is Tharsa, a red desert planet much like Mars: cold, dusty and airless, but rich in iron, metals and frozen water under the surface. The ark can land only once, and only if a beacon on the ground guides it through Tharsa's dust; until then, its small rockets can ferry passengers down.

One year short of orbit, a micrometeor storm damages the ark's reactor. The ship's computer wakes a landing crew of twelve and sends them down in the lander *Kestrel*. The remaining 188 colonists stay aboard in cryosleep. The ark's cryo systems still hold, but without guidance from the ground it can never land.

There is no base waiting for them. The crew must build one from nothing: mines, power, a sealed base, food — and a Landing Pad where the ark's rocket can bring down more hands. Their final goal is the Haven Beacon, a huge guidance tower built from Aurite, a rare alloy refined from Tharsa's steel and copper. Once it stands and at least 100 colonists on the surface are Healthy, the *Meridian* can finally make its guided fall to the surface and land safely.

2. Gameplay

2.1 Mechanics overview

The game rests on two systems: colonists with needs, and mines that only produce while colonists work in them. The player balances the two — deciding how many people go down the mines and how many keep the colony fed, built and healthy.

Colonist needs (common to all colonists)

Needs drain over time, and faster when working and carrying resources. Running out of Hunger, Thirst, Oxygen or Rest is fatal, and Health has three states: Healthy, Injured and Dead. The full behaviour for each attribute is in 3.1 Character. Needs are not shown on the character model: clicking a colonist opens an info panel with five need bars and a separate Health indicator (Healthy, Injured or Dead).

Need	Refilled by	Key rule
Hunger	Eating vegetables from storage	Stops working at 20 to look for food; dies at 0
Thirst	Drinking from the base's water supply	Stops working at 20 to look for water; dies at 0
Oxygen	Being inside the base while operational Oxygenators cover everyone inside, or working in a powered mine; suit tanks refill on entry at a powered Exit Door with an operational oxygen supply	Dies at 0
Rest	Sleeping in a Sleeping Room	At 20, seeks a bed; at 10, becomes exhausted and needs Hospital treatment by one Medic until Rest returns to 100; dies at 0
Happiness	Free time in the Arcade	Affects the rest as a multiplier
Health	Treatment at the Hospital	Healthy (100), Injured (50) or Dead (0)

Injuries. Miners working in a mine, and anyone outside during a dust storm, have a random chance of becoming Injured (Health 50). An Injured colonist cannot work or carry resources and goes to the Hospital to be healed. Rotating Miners and timing outdoor work between storms is part of the strategy.

Resources and mines

Only two resources are mined: iron and copper, each from its own deposit type and mine. They are processed into the two building materials, Steel and Components, which can be refined further into Aurite, used only for the Haven Beacon. Water is pumped automatically, and food and medicine are grown. Raw ores cannot be used directly for construction, repairs or robot manufacturing: they must first be processed into Steel or Components.

Resource	Comes from	Processed into	Used for
Iron ore	Iron Mine on an iron deposit (staffed)	Steel (Iron Processor)	Input for Steel production; must be processed before use
Copper ore	Copper Mine on a copper deposit (staffed)	Components (Copper Processor)	Input for Components production; must be processed before use
Steel	Iron ore processed in an Iron Processor (Miner + power)	Aurite (combined with Components in an Aurite Processor)	Construction material; also an input for Aurite production
Components	Copper ore processed in a Copper Processor (Miner + power)	Aurite (combined with Steel in an Aurite Processor)	Advanced buildings, Scout Robot manufacture and repairs to all buildings and structures; also an input for Aurite production
Vegetables	Greenhouse	—	Food
Medicinal plants	Greenhouse	—	Healing at the Hospital
Aurite	Aurite Processor (Steel + Components)	—	Only the Haven Beacon

Utilities. Electricity, water and oxygen are not items carried to Storage; they flow through the base's power grid, pipes and air supply.

Utility	Produced by	Stored in	Used by	If it runs out
Electricity	Solar Panels, Wind Turbines	Energy Batteries	All buildings	Buildings without power stop working
Water	Water Extractor (automatic, needs electricity)	Water Deposits (surplus water; separate from Storage)	Colonists drinking, Greenhouses, Oxygenators	Thirst can't be refilled; Greenhouses stop growing; Oxygenators stop producing oxygen

Utility	Produced by	Stored in	Used by	If it runs out
Oxygen	Oxygenators (inside the base, up to 25 people each); mines (for their own Miners); both use electricity	Suit tanks only, refilled on entry at a powered Exit Door with an operational oxygen supply	Colonists inside the base, Miners at work, anyone outside on suit tanks	Colonists' Oxygen drains; at 0, they die

Each Greenhouse has several planting slots. Clicking a Greenhouse opens a panel showing every slot, and the player picks what to plant in each one: **vegetables**, which colonists eat straight from Storage, or **medicinal plants**, which Medics use to heal patients at the Hospital. Balancing the two across all slots is a constant trade-off.

How a mine works

- **Must be staffed.** Each mine has 5 Miner slots. With nobody inside it produces nothing; output grows with every Miner working there. Miners go to mines on their own, following building priorities. Miners don't rely on their suit tanks while working: the mine supplies them with oxygen automatically.
- **Needs Power.** It must be connected to the power grid. Without power, it stops producing ore and oxygen, and its Miners have to switch to their suit tanks and head back.
- **Output bin.** Ore piles up in the mine's own bin (20 units). Any healthy, available colonist can carry it inside through the Exit Door to Storage or processors, after urgent needs and job-specific tasks; a full bin pauses the mine.
- **Finite deposits.** Each deposit shows how much is left. The richest deposits lie far from the base, so trips outside get longer and riskier.

Buildings

The colony is split into a sealed **interior base**, where colonists live and breathe, and the open **exterior**, where power, water and ore come from. The Exit Door is the only way between the two: every trip outside, and every load of ore carried in, passes through it. Unless noted, buildings work on their own as long as they have power. All buildings and structures gradually accumulate damage through wear over time. During dust storms, all buildings and structures have a higher chance of suffering additional damage of a random amount; this is not limited to Solar Panels. Engineers repair accumulated damage by consuming Components made from copper ore.

Interior base

Building	Function
Exit Door	The only passage between inside and outside. Colonists suit up here and refill their suit oxygen on entry, provided the Exit Door has power and an operational oxygen supply
Oxygenator	Supplies breathable air for up to 25 people inside the base at the same time. Requires power and water; without either it stops producing oxygen. If operational Oxygenators cannot cover everyone inside, everyone's Oxygen starts to drain
Iron Processor	Turns iron ore into Steel; requires a Miner and power
Copper Processor	Turns copper ore into Components; requires a Miner and power
Aurite Processor	Refines Steel and Components into Aurite; requires a Miner and power
Greenhouse	Has several planting slots; clicking it opens a panel where the player chooses vegetables or medicinal plants for each slot. Needs a Farmer, water and power
Sleeping Room	Beds where colonists restore Rest without a Medic; colonists seek a bed at Rest 20
Arcade	Recreation room where colonists restore Happiness in their free time
Storage	One shared store for every resource up to 100; each extra Storage adds capacity
Hospital	Treats injuries with medicinal plants and exhaustion when Rest reaches 10. Each patient requires one Medic during treatment; exhausted patients stay until Rest returns to 100. Normal sleep in a Sleeping Room does not require a Medic

Exterior

Building	Function
Solar Panel	Cheap main power source by day; nothing at night, very little in dust storms, and needs cleaning by Engineers afterwards
Wind Turbine	Steady power day and night; produces more during dust storms and strong winds
Energy Battery	Stores surplus power for the night and for storms
Iron Mine	Extracts iron ore from an iron deposit; needs Miners and power
Copper Mine	Extracts copper ore from a copper deposit; needs Miners and power
Water Extractor	Pumps a certain amount of water automatically; needs power only

Building	Function
Water Deposit	Stores surplus extracted water separately from the resources kept in Storage
Charge Pad	Where the Scout Robot recharges its battery
Landing Pad	Where the ark's rocket lands with new colonists
Radar	Provides advance warning of a dust storm
Haven Beacon	Final building; costs 300 Aurite. Victory requires both a completed Beacon and at least 100 Healthy colonists on the surface; colonists are not consumed

Jobs and robots. All colonists are the same except for their job. Any healthy, available colonist can transport ore after urgent needs and job-specific tasks have been addressed. Exploration is done by robots, not people.

Role	What they do
Engineer (job)	Builds every new building and repairs all damaged buildings and structures, consuming Components manufactured from copper ore in the Copper Processor. Also cleans dust off Solar Panels after storms. When no job-specific task is pending and urgent needs are met, can carry ore from mines to Storage or processors, like other healthy, available colonists
Miner (job)	Works in Iron or Copper Mines and operates Iron, Copper and Aurite Processors. Mines and processors produce nothing without Miners; Miners choose workplaces according to building priorities.
Farmer (job)	Runs the Greenhouses: vegetables and medicinal plants
Medic (job)	Treats Injured colonists at the Hospital with medicinal plants and helps exhausted patients recover Rest to 100.
Scout Robot (unit)	Drives out to reveal iron and copper deposits; its battery drains as it travels, and it must return to a Charge Pad to recharge before the next trip.

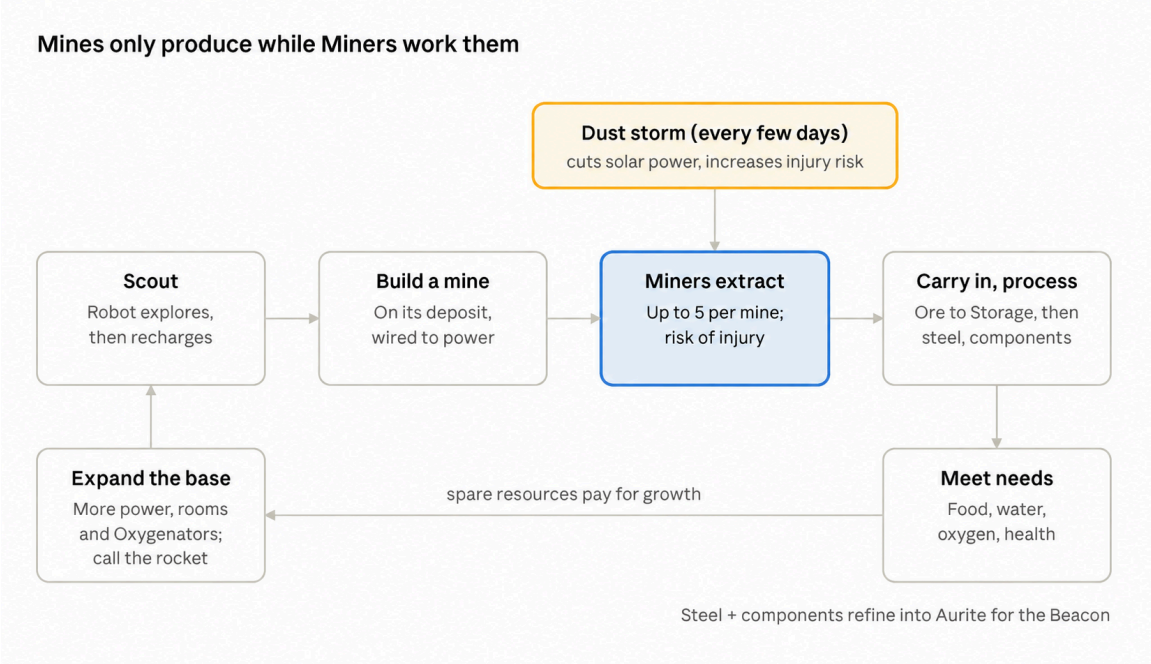
Other systems

- **Landing Pad and rockets.** The only way to grow the population. The ark originally carries 200 colonists: 12 form the initial landing crew and 188 remain aboard. Once a Landing Pad is built, the player can call the rocket at no cost while passengers remain aboard. Each flight brings up to 5 colonists, drops them off and flies back up on its own; the final flight may bring fewer. It can be called again once it has returned to the ark. Deaths do not replenish the passenger pool.
- **Oxygen and the base.** Inside the base, colonists breathe as long as Oxygenators with power and water provide enough capacity for everyone indoors. Powered mines maintain their working

Miners' Oxygen through their own supply, which depends on electricity rather than the base's water supply. Anywhere else outside, colonists use limited suit oxygen. Suit tanks refill only on entry at an Exit Door with power and an operational oxygen supply.

- **Power.** Solar Panels and Wind Turbines feed one grid, with Energy Batteries covering the gaps. Losing power stops all buildings.
- **Dust storms.** Every few in-game days a dust storm hits. Solar output drops sharply while Wind Turbines produce more; mines clog and need an Engineer, and anyone outside burns oxygen faster and risks injury. All buildings and structures also have a higher chance of taking additional damage of a random amount during the storm, on top of their gradual wear over time. Engineers repair this damage using Components made from copper ore. The player can declare an emergency that prevents colonists from going outside and orders those already outside to return to the base.
- **Building priorities.** Clicking any building opens a popup with its status and a priority setting (Low, Normal, High). Colonists act on their own but serve higher-priority buildings first: Miners staff the highest-priority mines and processors, Engineers build and repair high-priority buildings first, Farmers tend high-priority Greenhouses, and healthy, available colonists carry ore from high-priority mines first, after urgent needs and job-specific tasks. Buildings have no upgrades: their design and capabilities do not change after construction; maintenance repairs accumulated damage.

2.2 Core loop



core loop · 6 steps, dust storm pressure

Minute to minute, the player keeps the loop running: find deposits, build mines, staff them with Miners, and turn the ore into what the colony needs. Dust storms interrupt the loop, increase the risk of injury

for anyone outside and punish colonies without battery storage. Each expansion reaches richer deposits, until the colony has enough Steel and Components to refine the Aurite for the Haven Beacon.

2.3 Win/Lose conditions

Win: complete the **Haven Beacon**, which costs 300 Aurite, and have at least 100 Healthy colonists on the surface. Victory triggers as soon as both conditions are met, regardless of their order; the colonists are not consumed. The *Meridian* then lands safely.

Lose: the only way to lose is for every colonist on the surface to die. There is no time limit.

Soft failure: a colonist dying is a setback, not a loss. It removes a worker and lowers everyone's Happiness.

Endless mode (optional): after the ark lands, keep playing to house and feed everyone it carried.

3. The 3 C's

3.1 Character

The player is the **Colony Director**, an unseen commander; the characters on screen are the human colonists, who act on their own.

Design of a colonist

- Humans in space suits both outside and inside.
- Slightly stylised proportions (larger heads and hands) so they read well from a high camera.
- Job shown by suit colour: Engineers in orange, Miners in yellow, Farmers in green, Medics in white.
- **Scout Robot:** a small six-wheeled rover.
- Needs and Health are not shown on the model. Clicking a colonist opens an info panel with their name, job and five need bars and a separate Health indicator (Healthy, Injured or Dead).

Colonist behaviour by attribute

Attribute	Behaviour
Hunger	At 20, the colonist stops working and looks for food. At 0, dies
Thirst	At 20, the colonist stops working and looks for water. At 0, dies
Oxygen	At 0, dies
Rest	At 20, stops working and seeks a bed in a Sleeping Room, where normal sleep restores Rest without a Medic. At 10, becomes exhausted and goes to the Hospital, taking up one Medic until Rest returns to 100. At 0, dies
Happiness	Affects the Rest stat
Health	Three states: Healthy (100), Injured (50) and Dead (0). Miners in mines and anyone outside during a dust storm have a random chance of becoming Injured.

Attribute	Behaviour
	An Injured colonist cannot work or carry resources and goes to the Hospital to be healed

3.2 Camera

A free 3D strategy camera, as in city builders.

- Default view looks down at about 50°, zoomed out enough to show the base and the nearest mines. When the camera zooms in over the base, its roof fades out to show the rooms and the colonists inside.
- The player can pan, rotate around the focus point and zoom from a wide view of the map down to a close view of one colonist.
- During a dust storm the camera adds blowing red dust and lower visibility, but stays fully controllable.

3.3 Control

Control is indirect: the player gives orders to the colony, not to one character. All input is mouse and keyboard.

Action	Input
Move camera	WASD or move the mouse to the screen edge
Rotate camera	Q / E or hold middle mouse button
Zoom	Mouse wheel
Select colonist, robot or building (opens its popup; a building's popup includes its priority setting and, for Greenhouses, the planting slots)	Left click
Cancel construction / close menu	Right click cancels construction or closes an open menu; Esc cancels or closes the current selection/menu
Send Scout Robot to explore	Select the robot, then right click a point on the map with no menu or construction mode open
Build menu (Interior and Exterior tabs)	B, then left click to place, R to rotate
Call the rocket from the ark	Select the Landing Pad, press L
Pause / game speed	-/+
Clear emergency / declare emergency	F1 / F2, respectively

The player decides **what** happens (build an Iron Mine on this deposit, this colonist is a Farmer, food before comfort); colonists decide **how** and **when**, based on their needs and job.

4. Game and visual references

Game	What it shares with Arkfall	What we do differently
Surviving Mars	Colonising a Mars-like planet; domes, rockets bringing colonists from orbit	One clear goal: building the Haven Beacon
Planetbase	Oxygen-based survival on an alien planet; landing pad for new arrivals; jobs	Mines tied to specific deposits, with injury risk for Miners




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35
Trading Ship has landed